

HYPERLIQUID

Data Analytics Report

Trading Behaviour Analysis

Trader behaviour and performance across
Fear, Neutral and Greed market sentiment



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SECTION 01

Background

Hyperliquid is a cryptocurrency trading platform where users can buy and sell digital assets and trade on their expected price movements. It provides both **Spot** and **Perpetual** trading.

SPOT TRADING

Traders directly buy or sell a cryptocurrency at its current market price. For example, buying Bitcoin means the trader actually acquires Bitcoin and can later sell it when they choose.

PERPETUAL TRADING

Traders do not directly buy the cryptocurrency. Instead, they take a position based on whether they expect its price to increase or decrease. A **Long** position is taken when they expect the price to rise and a **Short** position when they expect the price to fall. Unlike regular futures contracts, perpetual trades do not have a fixed expiry date.

SECTION 02

Business Problem

Crypto traders operate under changing market sentiment, ranging from Fear to Greed. However, it is not clear whether traders change their trading behavior or experience different outcomes under different sentiment conditions.

This analysis aims to understand:

- How does market sentiment relate to trader profitability?
- Does trading activity and trade size change across Fear, Neutral, and Greed conditions?
- Does trader risk-taking behavior differ across sentiment conditions?
- Are traders more successful when their trading behavior aligns with or diverges from market sentiment?
- Are there behavioral patterns associated with better or worse trading outcomes?

The goal is not to predict the market, but to identify meaningful relationships between market sentiment, trader behavior, risk, and trading outcomes to better understand how traders behave under different market conditions.

IN SIMPLE WORDS

We want to answer one main question: **“When the market is Fearful, Neutral, or Greedy, how does trader behavior change, and does that behavior relate to different trading outcomes?”**



SECTION 04

Datasets

This project uses two complementary datasets to study the relationship between market sentiment and trader behavior.

The first dataset contains daily Bitcoin market sentiment, which describes whether the overall market sentiment was Fear, Neutral, or Greed on a particular day. The second dataset contains individual trade-level data from Hyperliquid, providing information about traders, their trades, position direction, trade size, and realized profit or loss. By combining these two datasets based on date, we can examine how trading behavior and performance vary under different market sentiment conditions.

Bitcoin Market Sentiment Dataset

The dataset contains **2,644 records** across **4 columns**, with a daily granularity, meaning each record represents the market sentiment recorded for one day.

Column	Data Type	Description	Example
timestamp	Integer	Unix timestamp representing the date	1733961600
value	Integer	Bitcoin Fear & Greed score ranging from 0 to 100	75
classification	String	Market sentiment category based on the sentiment score	Fear, Neutral, Greed
date	String	Human-readable date corresponding to the sentiment record	2024-12-12

Table 1. Bitcoin Market Sentiment dataset — column details.

Hyperliquid Trading Data

The dataset contains **211,224 records** across **16 columns**, with individual trade-level granularity, meaning each row represents a single trade executed on the Hyperliquid platform.

Column	Data Type	Description	Example
Account	String	Unique trader wallet address	0x1234...abcd
Coin	String	Cryptocurrency being traded	BTC, ETH, SOL
Execution Price	Float	Price at which the trade was executed	43250.50
Size Tokens	Float	Number of cryptocurrency tokens traded	0.5
Size USD	Float	Dollar value of the trade	21625.25
Side	String	Whether the trade was a Buy or Sell	Buy
Timestamp IST	String	Date and time of the trade in IST	2024-01-15 14:30:22
Start Position	Float	Position size before the trade	1.2



Column	Data Type	Description	Example
Direction	String	Direction or type of position action	Open, Close
Closed PnL	Float	Profit or loss realized from the trade	150.75, -45.20
Transaction Hash	String	Blockchain transaction identifier	0xabc123...
Order ID	Integer	Unique identifier for the order	123456
Crossed	Boolean	Indicates whether the order crossed the spread	True / False
Fee	Float	Trading fee associated with the trade	2.50
Trade ID	Float	Unique identifier for the trade	789012.0
Timestamp	Float	Unix timestamp representing the trade time	1705318822.0

Table 2. Hyperliquid trading dataset — column details.

SECTION 05

Methodology

Data Extraction

The datasets were loaded into Python using the Pandas library. After loading the data, an initial inspection was performed to understand the structure and quality of the datasets. The following steps were carried out:

Load the datasets

The Bitcoin Market Sentiment and Hyperliquid Trading datasets were loaded into Pandas DataFrames for further analysis.

Check for null values

The datasets were checked for missing or null values across all columns. This step was performed to identify incomplete records that could affect subsequent analysis.

Data Wrangling

After extracting the datasets, several data-wrangling steps were performed to prepare the data for analysis.

Convert date data type

The *date* column in both DataFrames was initially stored as a string. It was converted into the Pandas Datetime format to enable date-based operations and to facilitate joining the datasets.

Join the datasets

The Bitcoin Market Sentiment DataFrame and Hyperliquid Trading DataFrame were joined using the *date* column. This allowed the daily market sentiment to be associated with each corresponding trade.

Remove unnecessary columns

After joining the datasets, columns that were not required for the analysis were removed. These included:



- timestamp
- Order ID
- Transaction Hash
- Trade ID
- Timestamp

These columns were excluded because they were not required for the behavioral and performance analysis.

Group sentiment categories

The original sentiment dataset contained five categories: Extreme Fear, Fear, Neutral, Greed, and Extreme Greed. These were grouped into three broader categories:

Original category	Grouped category
Extreme Fear	Fear
Fear	Fear
Neutral	Neutral
Greed	Greed
Extreme Greed	Greed

Table 3. Sentiment category grouping.

Classify trading type

A new *trade_type* column was created based on the *Direction* column. Buy and Sell directions were classified as **Spot**, while other directions such as Close Long and Open Short were classified as **Perpetual**. This classification allows the analysis to compare trading behavior between Spot and Perpetual trades.



SECTION 06

Exploratory Data Analysis

EDA was performed to understand the distribution, relationships, and patterns within the trading data. The analysis was divided into three stages: **Univariate Analysis**, **Bivariate Analysis**, and **Multivariate Analysis**.

Univariate Analysis

Univariate analysis was performed to examine individual variables independently and understand their distribution and characteristics. Appropriate visualizations were selected based on the type of variable and the question being analyzed.

1. Distribution of Market Sentiment Classification

WHY THIS CHART WAS CHOSEN

It is important to understand the distribution of the dependent variable before proceeding with further analysis. A **pie chart** was chosen because it clearly represents the relative proportion and percentage of each sentiment category in the dataset.

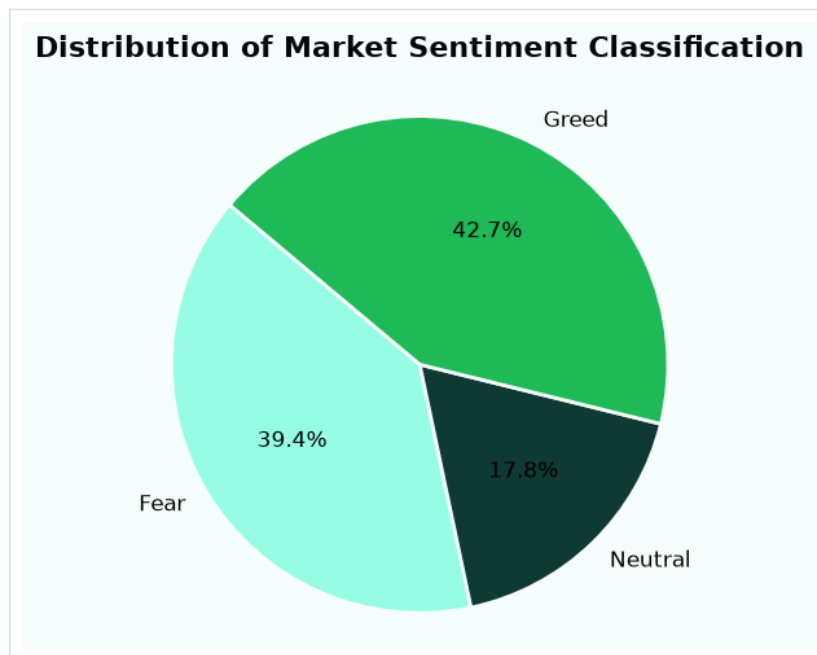


Figure 1. Distribution of market sentiment classification.

OBSERVATION

The market sentiment distribution consists of **42.7% Greed**, **39.4% Fear**, and **17.8% Neutral**. Greed represents the largest proportion of the observations, followed by Fear, while Neutral accounts for the smallest proportion. Overall, the dataset contains a relatively larger share of Fear and Greed observations compared with Neutral observations.



2. Distribution of Trade Size (USD)

WHY THIS CHART WAS CHOSEN

Size USD represents the **notional value of a trade**, or the total USD value involved in a single trade. Before analyzing how trade size changes across different conditions, it is important to understand the distribution of this continuous variable, including its typical values, spread, and range. A **histogram** was therefore chosen because it is suitable for visualizing the distribution of a continuous variable and helps identify the concentration, spread, and skewness of the trade sizes. The mean, median, and percentiles were also calculated to provide numerical measures of the distribution.

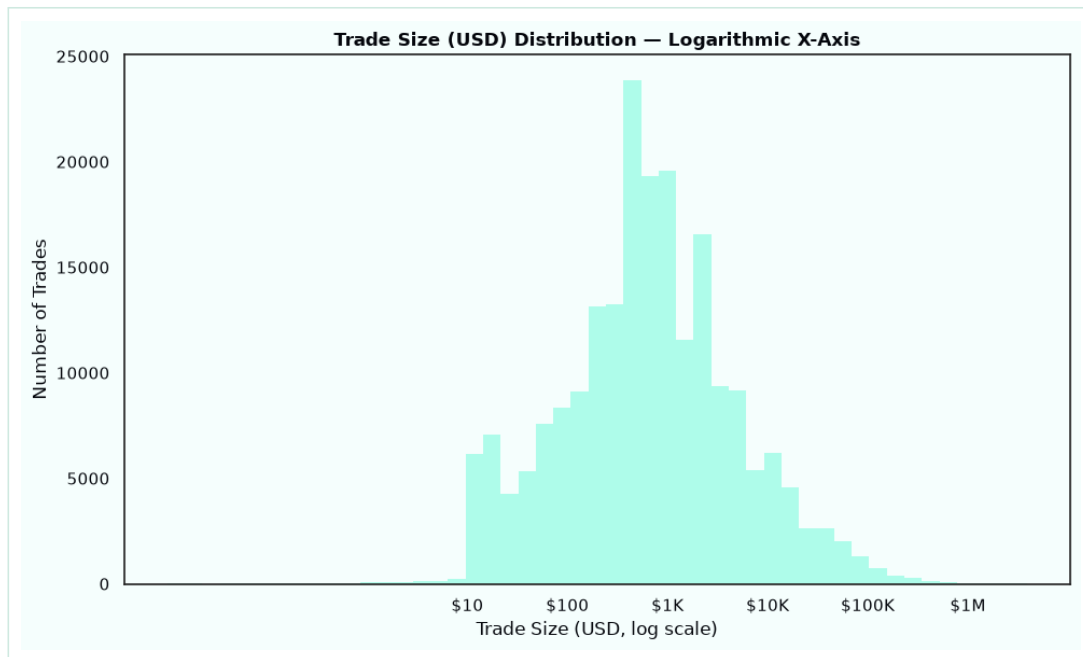


Figure 2. Trade size (USD) distribution on a logarithmic x-axis.

Statistic	Value
Median	\$597.28
Mean	\$5,636.17
Standard Deviation	\$36,588.16
25th Percentile (Q1)	\$194.00
75th Percentile (Q3)	\$2,058.51
95th Percentile	\$20,012.74
99th Percentile	\$88,887.23
Maximum	\$3,921,430.72

Table 4. Summary statistics for trade size (USD).



OBSERVATION

The distribution of trade size is strongly right-skewed. The median trade size is \$597.28, while the mean is considerably higher at \$5,636.17, indicating that a relatively small number of large trades are increasing the mean. The middle 50% of trades fall between \$194.00 (25th percentile) and \$2,058.51 (75th percentile).

The distribution becomes increasingly spread out at higher values, with the 95th percentile at \$20,012.74 and the 99th percentile at \$88,887.23. The maximum trade size reaches \$3.92 million, showing the presence of a small number of extremely large trades. The standard deviation of \$36,588.16, which is much larger than the median, further indicates substantial variability in trade sizes.

3. Distribution of Realized PnL

DATA PREPARATION

The *Closed PnL* column contains some records with a value of 0, particularly for trades where a position was opened or the trade did not yet result in a realized profit or loss. Since these records do not represent an actual realized outcome, trades with Closed PnL = 0 were excluded before plotting the realized PnL distribution.

WHY THIS CHART WAS CHOSEN

Understanding the distribution of Realized PnL is important for analyzing trading profitability and examining how profitability varies across different market sentiment conditions. Since Realized PnL is a continuous variable, a **histogram** was chosen to visualize the distribution of profits and losses, including their concentration, spread, and extreme values.

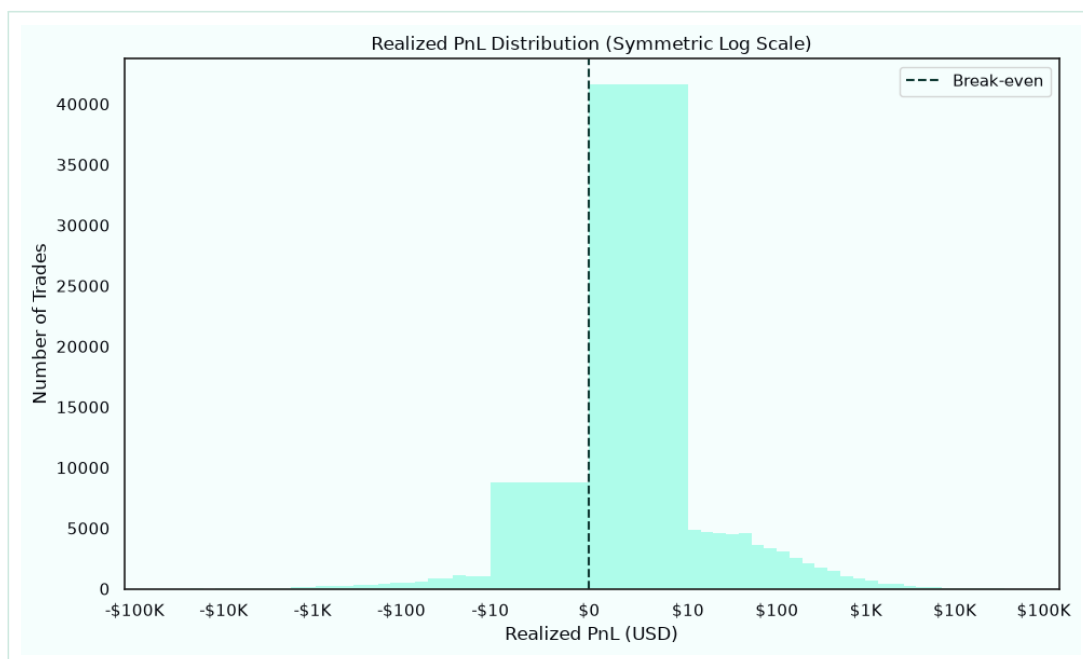


Figure 3. Realized PnL distribution on a symmetric log scale.



Statistic	Value
Mean	\$97.80
Median (50th Percentile)	\$6.05
Standard Deviation	\$1,299.45
Minimum	-\$117,990.10
25th Percentile (Q1)	\$0.41
75th Percentile (Q3)	\$38.18
90th Percentile	\$168.03
95th Percentile	\$392.68
99th Percentile	\$1,965.34
Maximum	\$135,329.09
Skewness	21.85
Kurtosis	3,225.92

Table 5. Summary statistics for realized PnL.

OBSERVATION

The realized PnL distribution is highly right-skewed, with most realized PnL values concentrated close to zero and a smaller number of trades generating substantially larger profits. The median PnL is \$6.05, while the mean is considerably higher at \$97.80, indicating that large profitable trades have a strong influence on the average. The middle 50% of realized PnL values lie between \$0.41 and \$38.18, while the 95th percentile is \$392.68 and the 99th percentile reaches \$1,965.34.

At the same time, the distribution contains both substantial losses and large profits, with a minimum of -\$117,990.10 and a maximum of \$135,329.09. The very high skewness (21.85) and kurtosis (3,225.92) further indicate that the distribution is heavily influenced by extreme values. The symmetric logarithmic scale in the chart helps visualize both the negative and positive PnL values across a wide range while keeping the break-even point at zero visible.



4. Profitable Trades vs. Loss Trades

WHY THIS CHART WAS CHOSEN

Understanding the number of profitable and loss-making trades is important for evaluating the overall trading outcomes and provides a basis for calculating and comparing the **win rate across different market sentiment conditions** in the later analysis. A **bar chart** was chosen because it provides a clear visual comparison of the number of trades in each category, making the difference between profitable and loss trades easy to identify.

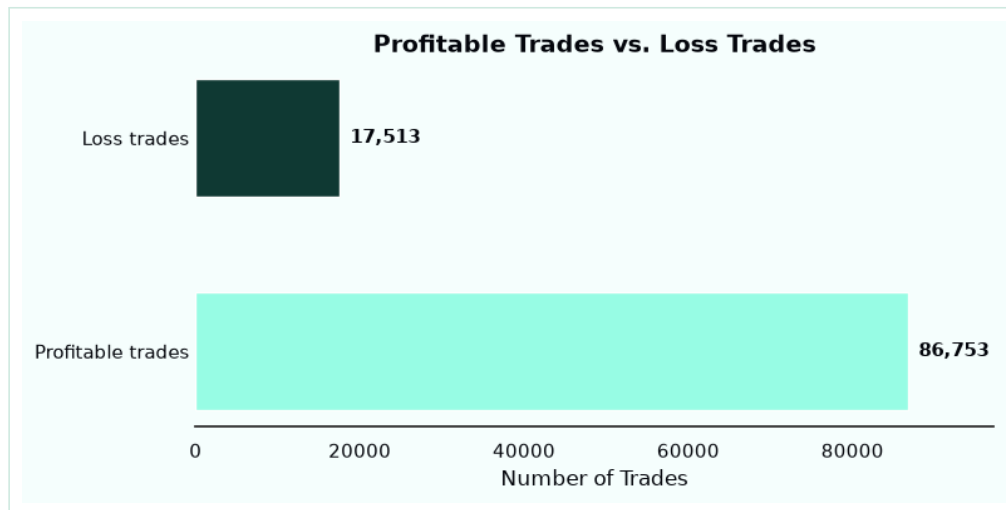


Figure 4. Number of profitable trades versus loss trades.

OBSERVATION

The dataset contains **86,753 profitable trades** compared with **17,513 loss trades**. Profitable trades therefore represent a substantially larger number of observations than loss trades in the analyzed dataset. This provides the overall distribution of trade outcomes, which will be further examined across **Fear, Neutral, and Greed** sentiment conditions in the bivariate analysis.

5. Number of Trades by Trading Type

WHY THIS CHART WAS CHOSEN

The trading data contains two main trading types: **Spot** and **Perpetual**. Comparing the number of trades in each category helps understand the type of trading that occurs more frequently in the dataset. A **bar chart** was chosen because it provides a direct and clear comparison of the number of trades between the two trading types.

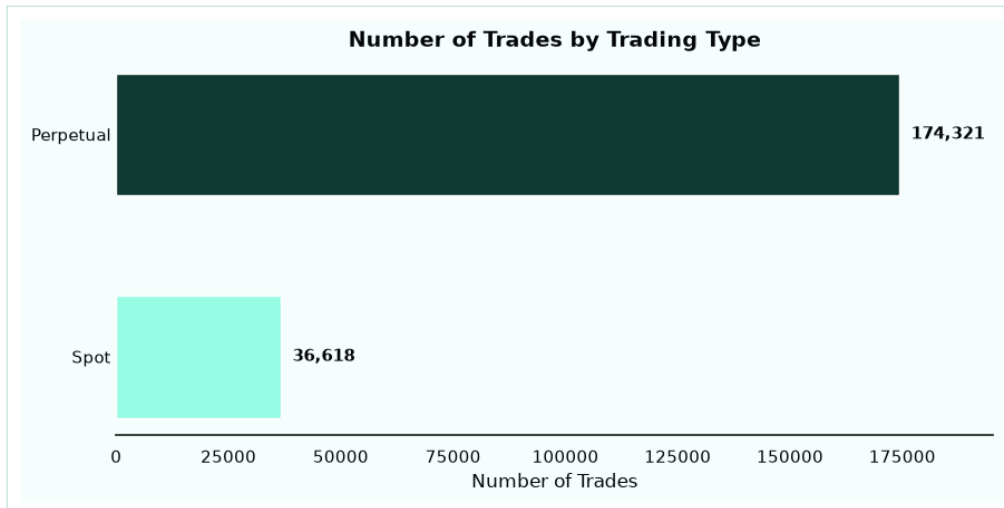


Figure 5. Number of trades by trading type.

OBSERVATION

The dataset contains **174,321 Perpetual trades** and **36,618 Spot trades**. Perpetual trading accounts for a substantially larger number of trades than Spot trading in the analyzed dataset. This indicates that Perpetual trades form the majority of the trading activity, while Spot trades represent a smaller portion. Further analysis can examine how these two trading types behave under different market sentiment conditions.

Bivariate Analysis

Bivariate analysis was performed to examine the relationship between **two variables** and understand how one variable varies across another.

1. Profitability vs. Market Sentiment

WHY THIS ANALYSIS WAS CHOSEN

One of the main objectives of this analysis is to understand how trading profitability varies under different market sentiment conditions. Therefore, **market sentiment** was compared with **realized profitability** across Fear, Neutral, and Greed conditions. To compare profitability, the **mean and median Realized PnL** were calculated for each sentiment category. The **win rate** was also calculated to compare the proportion of profitable realized trades across sentiment conditions. In addition, **box plots** were used to visualize the distribution and spread of Realized PnL within each sentiment category, including the presence of extreme values.

Market Sentiment	Mean Realized PnL	Median Realized PnL	Win Rate
Fear	\$101.76	\$6.35	84.43%
Neutral	\$71.27	\$4.58	82.40%
Greed	\$104.80	\$6.49	82.45%

Table 6. Realized PnL and win rate by market sentiment.

**OBSERVATION**

The mean realized PnL was \$101.76 during Fear, \$71.27 during Neutral, and \$104.80 during Greed. The median realized PnL was much lower in all three conditions, at \$6.35, \$4.58, and \$6.49, respectively. This difference between the mean and median indicates that the average PnL is influenced by larger trades and extreme outcomes.

The win rate was 84.43% during Fear, 82.40% during Neutral, and 82.45% during Greed. The win rates were therefore relatively close across the three sentiment conditions, with Fear showing a somewhat higher observed rate in this dataset.

Overall, the results show differences in the observed profitability measures across sentiment conditions. The box plot in the next section supports the numerical results by showing differences in the central tendency and spread of realized PnL across sentiment conditions.

2. Trade Size (USD) vs. Market Sentiment

WHY THIS ANALYSIS WAS CHOSEN

One of the main objectives of the analysis is to understand whether trading activity and trade volume vary under different market sentiment conditions. Since *Size USD* represents the notional value of each trade, it was used as a measure of trading volume. To compare trading activity across Fear, Neutral, and Greed, the **total trade volume**, **mean trade size**, and **median trade size** were calculated for each sentiment category. These measures provide different perspectives: total volume shows the overall trading activity, while the mean and median show the typical trade size within each sentiment condition.

Market Sentiment	Total Trade Volume	Mean Trade Size	Median Trade Size
Fear	\$596.77M	\$7,177.14	\$749.58
Neutral	\$179.94M	\$4,779.42	\$547.31
Greed	\$412.18M	\$4,572.56	\$552.67

Table 7. Trade volume and trade size by market sentiment.

OBSERVATION

The total trade volume was highest during Fear, at approximately \$596.77 million, followed by Greed at \$412.18 million and Neutral at \$179.94 million. The mean trade size followed a similar pattern, with \$7,177.14 during Fear, compared with \$4,779.42 during Neutral and \$4,572.56 during Greed. The median trade size was also highest during Fear at \$749.58, while Neutral and Greed had relatively similar median values of \$547.31 and \$552.67, respectively.

The difference between the mean and median across all three sentiment conditions indicates that the trade-size distributions are influenced by relatively large trades. Therefore, the median provides a more representative view of the typical trade size, while total volume reflects the overall amount of trading activity.



3. Risk vs. Market Sentiment

WHY THIS ANALYSIS WAS CHOSEN

One of the objectives of the analysis is to understand how trading risk varies across different market sentiment conditions. To measure the variability of trading outcomes, the **standard deviation of Realized PnL** was calculated for Fear, Neutral, and Greed. Standard deviation indicates how widely the realized PnL values are dispersed around their mean, so a higher standard deviation represents greater variability in realized PnL. A **box plot** was also used to visualize the spread of Realized PnL under each sentiment condition.

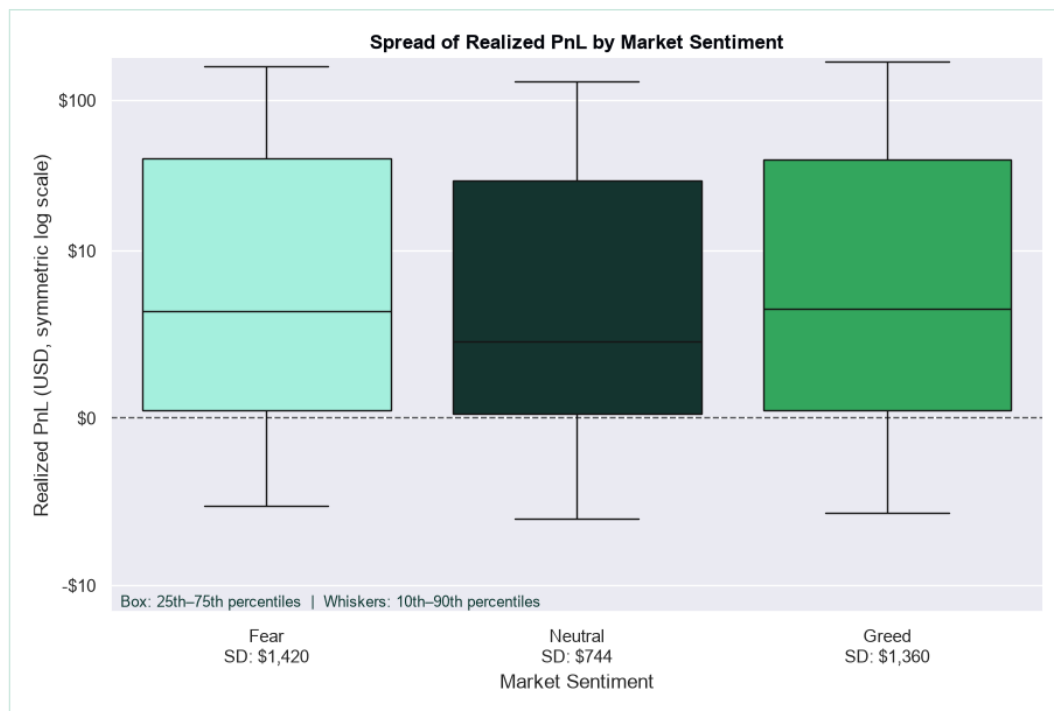


Figure 6. Spread of realized PnL by market sentiment.

OBSERVATION

The variability of realized PnL was highest during **Fear**, with a standard deviation of approximately **\$1,420**, followed by **Greed at \$1,360**. **Neutral** had a considerably lower standard deviation of approximately **\$744**. The box plot also shows a wider spread of PnL values under Fear and Greed compared with Neutral. This indicates that the realized PnL outcomes were more dispersed during Fear and Greed, while Neutral showed comparatively lower variability in the analyzed data.



4. Trade Direction vs. Market Sentiment

WHY THIS ANALYSIS WAS CHOSEN

One of the objectives of the analysis is to understand **how trader behavior changes under different market sentiment conditions**. The *Direction* variable provides information about the type of trading action taken by traders, such as opening or closing Long and Short positions, as well as Buy and Sell trades. To examine how these actions are distributed within each sentiment condition, the number of trades for each direction was calculated separately for **Fear**, **Neutral**, and **Greed**, and then converted into percentages within each sentiment category. A **pie chart** was used for each sentiment condition to clearly show the relative share of each trading direction.

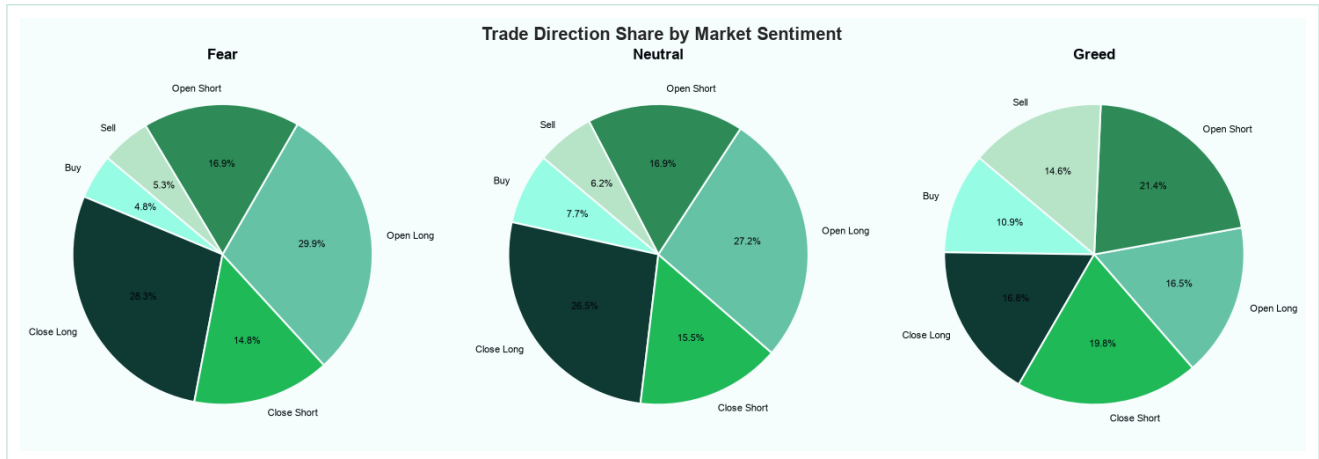


Figure 7. Trade direction share by market sentiment.

Market Sentiment	Open Long	Close Long	Open Short	Close Short	Buy	Sell
Fear	29.9%	28.3%	16.9%	14.8%	4.8%	5.3%
Neutral	27.2%	26.5%	16.9%	15.5%	7.7%	6.2%
Greed	16.5%	16.8%	21.4%	19.8%	10.9%	14.6%

Table 8. Share of each trade direction within each sentiment condition.

OBSERVATION

During **Fear**, Open Long (29.9%) and Close Long (28.3%) account for the largest shares of trading activity, followed by Open Short (16.9%) and Close Short (14.8%). Buy and Sell trades represent smaller shares at 4.8% and 5.3%, respectively.

During **Neutral**, Open Long (27.2%) and Close Long (26.5%) remain the largest categories. Open Short accounts for 16.9%, while Close Short represents 15.5%. Buy and Sell account for 7.7% and 6.2%, respectively.

During **Greed**, the distribution changes compared with Fear and Neutral. Open Short represents 21.4% and Close Short 19.8%, while Close Long accounts for 16.8% and Open Long 16.5%. Buy and Sell represent 10.9% and 14.6%, respectively.

Overall, the direction shares show that the composition of trading actions differs across sentiment conditions. **Open/Close Long actions form larger shares during Fear and Neutral, whereas Open/Close Short actions account for a larger combined share during Greed.**



Multivariate Analysis

Multivariate analysis extends the previous analysis by examining **three or more variables together**. While bivariate analysis helps us understand individual relationships such as **Profitability vs. Sentiment** and **Trade Size vs. Sentiment**, multivariate analysis allows us to examine how multiple aspects of trader behavior and trading outcomes interact under different market sentiment conditions.

1. Market Sentiment vs. Trading Type vs. Performance

WHY THIS ANALYSIS WAS CHOSEN

This analysis was performed to examine whether **trading performance varies across different market sentiment conditions and trading types**. Since the dataset contains two trading types — Spot and Perpetual — analyzing them separately helps identify whether the relationship between sentiment and performance differs between the two types of trading. The analysis compares **Fear, Neutral, and Greed** within each trading type using **median Realized PnL** and **win rate**. Box plots were used to visualize the distribution of Realized PnL, while the heatmap shows the win rate across the combinations of sentiment and trading type.

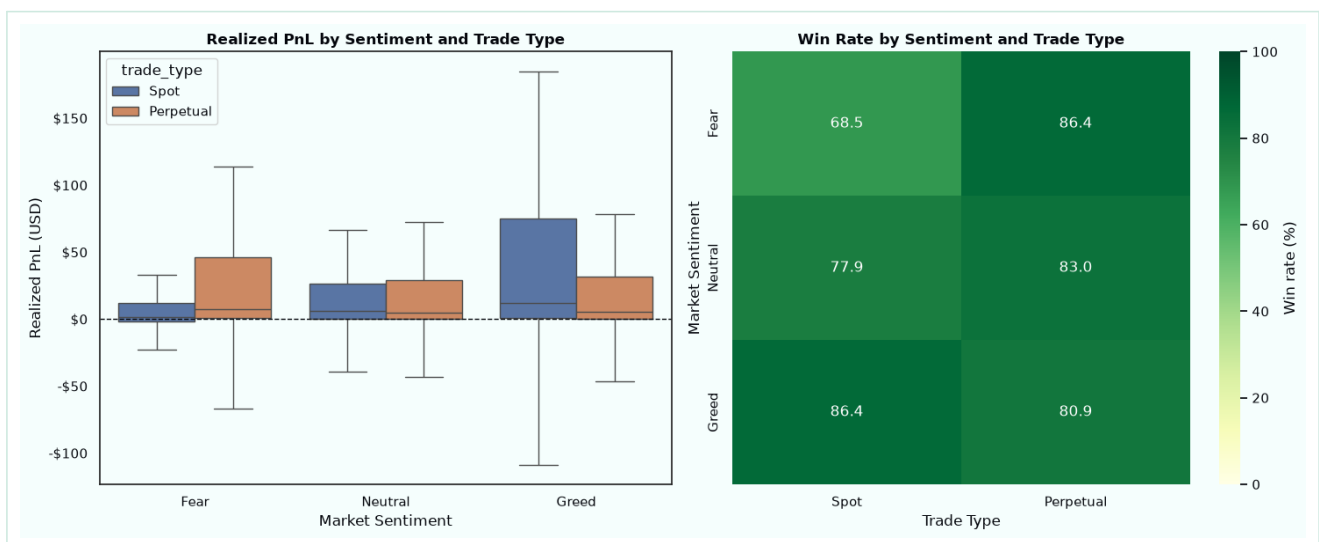


Figure 8. Realized PnL (box plot) and win rate (heatmap) by sentiment and trade type.

Market Sentiment	Spot — Median PnL	Spot — Win Rate	Perpetual — Median PnL	Perpetual — Win Rate
Fear	\$1.25	68.46%	\$7.33	86.36%
Neutral	\$5.72	77.89%	\$4.41	83.05%
Greed	\$12.13	86.44%	\$5.21	80.88%

Table 9. Median realized PnL and win rate by sentiment and trade type.

**OBSERVATION**

The observed median Realized PnL and win rates show differences between Spot and Perpetual trading across the three sentiment conditions.

Fear: Spot had a median Realized PnL of \$1.25 with a win rate of 68.46%, while Perpetual had a median of \$7.33 and a win rate of 86.36%.

Neutral: Spot had a median Realized PnL of \$5.72 and a win rate of 77.89%, while Perpetual had a median of \$4.41 and a win rate of 83.05%.

Greed: Spot had a median Realized PnL of \$12.13 with a win rate of 86.44%, while Perpetual had a median of \$5.21 and a win rate of 80.88%.

Across the sentiment conditions, the **Perpetual win rate was higher than Spot during Fear and Neutral**, while during **Greed, the Spot win rate was higher**. The median Realized PnL also varies across the combinations, showing that performance patterns are not identical between Spot and Perpetual trading under different sentiment conditions.

2. Position Direction vs. Market Sentiment vs. Realized PnL

WHY THIS ANALYSIS WAS CHOSEN

This analysis was performed to understand how **trading performance varies across different position directions under different market sentiment conditions**. Position direction indicates how a trade is related to a trader's position, such as Close Long, Close Short, or Sell. Since realized PnL is generated when positions are closed, examining these closing actions together with market sentiment helps identify whether realized PnL distributions differ across position directions and sentiment conditions. A **box plot** was used to compare the distribution, median, spread, and range of Realized PnL for each combination.

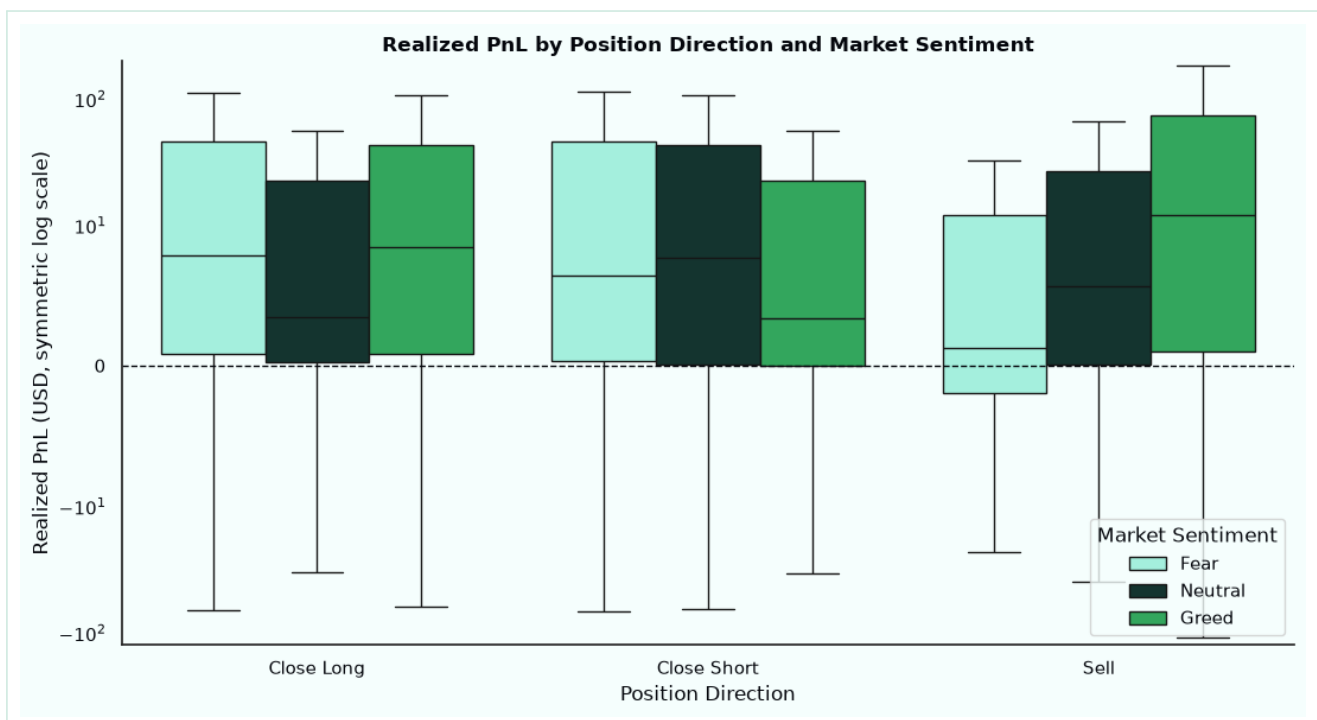


Figure 9. Realized PnL by position direction and market sentiment.

**OBSERVATION**

The distribution of Realized PnL varies across both position directions and sentiment conditions.

- For **Close Long**, the median Realized PnL is positive across all three sentiment conditions. The median is relatively higher during Greed and Fear than during Neutral.
- For **Close Short**, Fear and Neutral show higher median Realized PnL than Greed, while all three conditions show a wide range of positive and negative outcomes.
- For **Sell**, the median Realized PnL is comparatively low during Fear, higher during Neutral, and highest during Greed.

Across the chart, all position directions show considerable variation in Realized PnL, with outcomes extending on both sides of the break-even line. This indicates that position direction and market sentiment together are associated with differences in the distribution of realized trading outcomes in the analyzed data.



SECTION 07

Key Insights

1

Trading activity was different across market sentiment conditions

The distribution of market sentiment consisted of 42.7% Greed, 39.4% Fear, and 17.8% Neutral observations. Trade activity also differed across these conditions. Fear recorded the highest total trade volume at approximately \$596.77 million, followed by Greed at \$412.18 million and Neutral at \$179.94 million. The median trade size was also higher during Fear (\$749.58) compared with Neutral (\$547.31) and Greed (\$552.67).

This indicates that the size and overall volume of trading activity varied across sentiment conditions, with Fear showing larger observed trade activity.

2

Realized profitability varied across sentiment, but the differences were relatively close

The mean Realized PnL was \$101.76 during Fear, \$71.27 during Neutral, and \$104.80 during Greed, while the median was \$6.35, \$4.58, and \$6.49, respectively. The relatively large difference between mean and median shows that the average PnL is influenced by larger outcomes. The realized win rates were 84.43% for Fear, 82.40% for Neutral, and 82.45% for Greed.

Therefore, the analysis shows differences in profitability measures across sentiment conditions, but the win rates themselves remain relatively close.

3

Risk and variability were higher during Fear and Greed

Realized PnL showed substantially different levels of variability across sentiment conditions. The standard deviation was approximately \$1,420 during Fear, \$744 during Neutral, and \$1,360 during Greed. The box plots also show wider PnL distributions for Fear and Greed compared with Neutral.

This indicates that trading outcomes were more dispersed during Fear and Greed, meaning that the range of realized outcomes was wider under these conditions.

4

Trader direction changed across sentiment conditions

The distribution of trade directions showed differences in trading behavior across sentiment conditions. During Fear, Open Long (29.9%) and Close Long (28.3%) represented the largest shares. Neutral showed a similar pattern, with Open Long (27.2%) and Close Long (26.5%) being the largest categories. During Greed, Open Short (21.4%) and Close Short (19.8%) represented larger shares, while Open Long and Close Long accounted for 16.5% and 16.8%, respectively.

This shows that the composition of trading actions changed across sentiment conditions, particularly between Fear/Neutral and Greed.



5 Spot and Perpetual trading showed different performance patterns across sentiment

Perpetual trading represented the majority of the dataset, with 174,321 Perpetual trades compared with 36,618 Spot trades. When performance was examined jointly with sentiment, the observed patterns differed between the two trading types.

For Fear, Perpetual had a median Realized PnL of \$7.33 and an 86.36% win rate, compared with Spot's \$1.25 median and 68.46% win rate. During Neutral, Perpetual had a median of \$4.41 and an 83.05% win rate, compared with Spot's \$5.72 and 77.89%. During Greed, Spot had a median of \$12.13 and an 86.44% win rate, compared with Perpetual's \$5.21 and 80.88%.

Thus, the relationship between sentiment and observed performance was not identical for Spot and Perpetual trading.

6 Position direction and sentiment together showed differences in realized PnL distributions

The multivariate analysis of position direction, sentiment, and Realized PnL showed that the distribution of realized outcomes varied across combinations of these variables. Close Long, Close Short, and Sell displayed different median values and spreads under Fear, Neutral, and Greed. The distributions also extended on both sides of the break-even point, showing that both positive and negative realized outcomes occurred across the different combinations.

This indicates that market sentiment alone does not describe the full variation in trading outcomes; the type of trading action and position direction also provide additional context.

OVERALL KEY INSIGHT

Putting the findings together, the analysis shows that trader behavior, trading activity, risk, and realized performance varied across Fear, Neutral, and Greed conditions. Fear was associated with larger observed trade sizes and higher PnL variability, while Greed showed a different composition of trading directions and relatively high PnL variability. Profitability measures differed across sentiment conditions, but the observed win rates were relatively close. The multivariate analysis further showed that trading type and position direction added important differences to the observed performance patterns.

Most importantly, these results identify relationships and patterns in the analyzed data; they do not establish that market sentiment directly caused the changes in trader behavior or performance.

SECTION 08

Conclusion

This analysis examined how trader behavior, trading activity, risk, and trading performance vary across Fear, Neutral, and Greed market sentiment conditions. The results show that market sentiment was associated with noticeable differences in trading activity and the composition of trading actions.

Fear showed the highest total trade volume and median trade size, while Fear and Greed showed greater variability in realized PnL compared with Neutral. Profitability measures also varied across sentiment conditions, although the win rates remained relatively close. The analysis of trade direction showed differences in the types of actions taken under different sentiment conditions, with Long-related actions



representing larger shares during Fear and Neutral, while Short-related actions represented larger shares during Greed.

The analysis also showed that trading type provides additional context to performance. Spot and Perpetual trading displayed different median realized PnL and win-rate patterns across the three sentiment conditions. Similarly, examining position direction together with sentiment showed differences in the distribution of realized PnL.

Overall, the findings indicate that trader behavior and trading outcomes are not uniform across market sentiment conditions. Differences can be observed in trading activity, trade direction, risk, and performance, and these patterns become more detailed when trading type and position direction are considered together.

A NOTE ON INTERPRETATION

The analysis identifies associations and observed patterns rather than causal relationships, so the results should not be interpreted as evidence that market sentiment directly causes changes in trader behavior or profitability.